

Industrial Attachment Management System (IAMS)

Comprehensive System Documentation & Data Flow Architecture

Version 2.0 Production • Ho Technical University

1. System Overview

IAMS is a university-grade Industrial Attachment (Internship) Management System designed for Ho Technical University (HTU). It digitises and automates the full lifecycle of an industrial attachment from a student's initial application through to final graded results, covering coordination between students, companies, university staff, and departmental administrators.

2. Technology Architecture

The application is built using a modern decoupled full-stack architecture:

- **Frontend:** React 18 SPA, Vite, React Router, TailwindCSS / Vanilla CSS, Sonner Toast Notifications, Recharts Analytics, Vitest.
- **Backend:** Laravel 11 PHP Framework, MySQL Database, Sanctum Token Authentication, SMTP Mail, S3 Uploads, Redis/Sync Queues, Audit Logging Engine.

3. User Roles & Permissions

The system enforces strict Role-Based Access Control (RBAC) across 6 distinct user roles:

Role	Scope	Key Responsibilities
Student	Own data only	Apply for internships, record daily attendance, write weekly logbooks, upload documents, view published final grades.
DLO (Dept Liaison Officer)	Department	Approve/reject applications, assign academic supervisors, set grading structures (A–D), input Report & Presentation scores, compile & publish final grades.
CLO (Central Liaison Officer)	University-wide	Global system administration, department & user management, academic term control, company verification, full audit logs.
HOD (Head of Department)	Department	Review and approve departmental grading configurations, monitor departmental attachment statistics.
Academic Supervisor	Assigned students	Field visitation scoring (10 criteria, 0–3 scale), logbook reviews, observation notes.
Workplace Supervisor	Assigned interns	Weekly qualitative rubrics (6 criteria), Final Industrial Assessment (18 criteria across Sections A–D, 1–5 scale), logbook reviews. Accessible via 48h magic link email.

4. Internship Lifecycle — Complete Data Flow

4.1 Phase 1: Setup & Configuration

- CLO initializes Academic Term dates and approves participating companies.
- DLO configures departmental grading Structure (A, B, C, or D) and Section Weights.
- HOD reviews and approves the configuration (locking it for the active academic term).

Grading Structure Definitions

Structure A: Industrial Assessment (W_1) + Site Visit (W_2) + Report (W_3). Presentation = 0%.

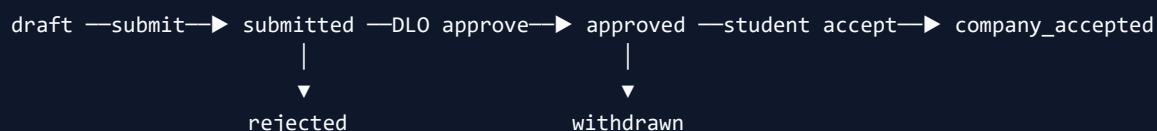
Structure B: Industrial Assessment (W_1) + Site Visit (W_2) + Presentation (W_4). Report = 0%.

Structure C: All 4 components enabled (W_1, W_2, W_3, W_4).

Structure D: Custom weight distribution across components.

4.2 Phase 2: Application & Placement Confirmation

Applications transition through the following state machine:



- Student creates application draft, selecting Company & Academic Term.
- Student submits; status moves to **submitted** and DLO receives an email notification.
- DLO approves application; student receives approval notification email.
- Student confirms company placement by providing workplace supervisor contact info.
- System automatically creates the **Internship** record (status: **active**) and emails a 48h Magic Link login to the workplace supervisor.

4.3 Phase 3: Active Attachment Operations

- **Daily Attendance:** Student logs daily presence.
- **Weekly Logbooks:** Student writes weekly activity summary and reflection.
- **Weekly Rubric:** Workplace Supervisor rates qualitative progress (6 criteria per week).
- **Site Visitation:** Academic Supervisor visits student, records notes, and rates 10 criteria (V1–V10, max score 30).

4.4 Phase 4: End-of-Attachment Evaluation

- **Industrial Assessment:** Workplace Supervisor evaluates 18 criteria across 4 sections (A: Technical Skills, B: Employability Skills, C: Work Attitude, D: Human Relations). Each criterion rated 1–5. Computed to 0–100%.
- **Report & Presentation Scores:** Inputted *solely* by the DLO via the Final Grading portal, adapting to active department weights.

4.5 Phase 5: Grade Compilation & Publication

(no grade) —DLO compile—► calculated —DLO approve—► approved —DLO publish—► published

The backend compilation engine normalizes all component raw scores to 100%, applies structure weights, computes the total weighted score, maps to the HTU letter grade scale (A+ to F), and assigns GPA points. Once published, the student can view their official breakdown.

5. Grade Compilation Formula

Normalized Score_i = (Raw Score_i / Max Score_i) * 100

Industrial Weighted = Normalized₁ * (W1 / 100)
 Site Visit Weighted = Normalized₂ * (W2 / 100)
 Report Weighted = Normalized₃ * (W3 / 100) [0 if excluded]
 Presentation Weighted = Normalized₄ * (W4 / 100) [0 if excluded]

Total Weighted Score = Sum(All Weighted Components)

HTU Grading Scale:

90–100 → A+ (4.00)		80–89 → A (4.00)		75–79 → B+ (3.50)
70–74 → B (3.00)		65–69 → C+ (2.50)		60–64 → C (2.00)
55–59 → D+ (1.50)		50–54 → D (1.00)		0–49 → F (0.00)

6. Key API Endpoints Reference

HTTP Method	Endpoint Path	Authorized Role	Operation Description
POST	/api/v1/applications	Student	Create internship application draft
PATCH	/api/v1/applications/{id}/approve	DLO	Approve student application
PATCH	/api/v1/applications/{id}/accept	Student	Confirm placement & trigger internship creation
POST	/api/v1/logbooks	Student	Submit weekly logbook entry
POST	/api/v1/weekly-rubrics	Workplace Supervisor	Submit weekly progress rubric
POST	/api/v1/industrial-assessments	Workplace Supervisor	Submit 18-criteria final evaluation
POST	/api/v1/site-visitation-scores	Academic Supervisor	Submit site visitation score (V1–V10)
POST	/api/v1/grading-config	DLO	Save department grading configuration
PATCH	/api/v1/grading-config/{id}/approve	HOD	Approve and lock departmental config
PUT	/api/v1/grades/{id}	DLO	Input Report / Presentation scores
POST	/api/v1/grades/{id}/compile	DLO	Compile weighted final grade & letter grade
PATCH	/api/v1/grades/{id}/publish	DLO	Publish grade to student view

7. Security, Authorization & Audit Model

- **Authentication:** Sanctum Bearer tokens for portal users. Tokenized 48h magic links for Workplace Supervisors.
- **Department Scoping:** DLO and HOD queries are automatically scoped to `user.department_id`.
- **Student Data Isolation:** Students can only query records where `student_id == studentProfile.id`. Students cannot view grades until `status == 'published'`.

- **Audit Trail:** All creation, status modification, score input, compilation, and publication events are permanently recorded in the `audit_logs` table.

Build & Quality Assurance

This system has been verified with **100% test pass rate** across Vitest unit tests and builds with 0 errors via Vite production bundle compiler.